

TASK 3: SUPPLIER AND INVENTORY MANAGEMENT SYSTEM

STUDENT DETAILS

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- **Department:** B.Sc Computer Science With AI
- **Project Title:** THE ARTISAN CRUMB: Bakery & Pastry Management System

1. Scenario Context

To maintain daily production without delays or batch shortages, "THE ARTISAN CRUMB" closely tracks its ingredient suppliers (such as regional flour mills, dairy farms, and sugar refineries) alongside batch stock levels. The inventory management module links each finished bakery product directly to its primary supplier, maintaining real-time visibility into stock availability, threshold alerts, and out-of-stock items.

2. Table Design (DDL)

Create Supplier Base Table

```
SQL> CREATE TABLE Supplier (  
2     Supplier_ID  VARCHAR2(7),  
3     Supplier_Name VARCHAR2(30),  
4     Email        VARCHAR2(30),  
5     Phone        NUMBER(10),  
6     Address      VARCHAR2(50)  
7 );
```

Table created.

Create Inventory Base Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID  VARCHAR2(7),  
3     Product_ID    VARCHAR2(7),  
4     Supplier_ID   VARCHAR2(7),  
5     Stock_Quantity NUMBER,  
6     Stock_Status  VARCHAR2(15),  
7     Last_Updated  DATE  
8 );
```

Table created.

3. Relationships & Constraints

Primary Key Constraints

```
SQL> ALTER TABLE Supplier
2      ADD CONSTRAINT pk_supplier PRIMARY KEY (Supplier_ID);
```

Table altered.

```
SQL> ALTER TABLE Inventory
2      ADD CONSTRAINT pk_inventory PRIMARY KEY (Inventory_ID);
```

Table altered.

Foreign Key Constraints linking Inventory to Product and Supplier tables

```
SQL> ALTER TABLE Inventory
2      ADD CONSTRAINT fk_inv_product FOREIGN KEY (Product_ID)
3      REFERENCES Product(Product_ID);
```

Table altered.

```
SQL> ALTER TABLE Inventory
2      ADD CONSTRAINT fk_inv_supplier FOREIGN KEY (Supplier_ID)
3      REFERENCES Supplier(Supplier_ID);
```

Table altered.

Column Level Constraints on Supplier Table

```
SQL> ALTER TABLE Supplier
2      MODIFY Email VARCHAR2(30) NOT NULL;
```

Table altered.

```
SQL> ALTER TABLE Supplier
2      ADD CONSTRAINT uk_supplier_email UNIQUE (Email);
```

Table altered.

Domain Integrity CHECK Constraints on Inventory Table

```
SQL> ALTER TABLE Inventory
2      ADD CONSTRAINT chk_stock_qty CHECK (Stock_Quantity >= 0);
```

Table altered.

```
SQL>
SQL> ALTER TABLE Inventory
2      ADD CONSTRAINT chk_stock_status CHECK (Stock_Status IN ('Available', 'Out of Stock'));
```

Table altered.

4. Insert Values & Data Population

Interactive SQL Insertion Template for Supplier Table

```
SQL> INSERT INTO Supplier (Supplier_ID, Supplier_Name, Email, Phone, Address)
  2 VALUES ('&sup_id', '&sup_name', '&email', &phone, '&address');
Enter value for sup_id: SUP-001
Enter value for sup_name: Golden Wheat Mills
Enter value for email: info@goldenwheat.com
Enter value for phone: 9876543210
Enter value for address: 12 Industrial Estate, Chennai
old 2: VALUES ('&sup_id', '&sup_name', '&email', &phone, '&address')
new 2: VALUES ('SUP-001', 'Golden Wheat Mills', 'info@goldenwheat.com', 9876543210, '12 Industrial Estate, Chennai')

1 row created.

SQL> /
Enter value for sup_id: SUP-002
Enter value for sup_name: Dairy Fresh Farms
Enter value for email: contact@dairyfresh.com
Enter value for phone: 9765432109
Enter value for address: 45 Green Valley, Coimbatore
old 2: VALUES ('&sup_id', '&sup_name', '&email', &phone, '&address')
new 2: VALUES ('SUP-002', 'Dairy Fresh Farms', 'contact@dairyfresh.com', 9765432109, '45 Green Valley, Coimbatore')

1 row created.

SQL> /
Enter value for sup_id: SUP-003
Enter value for sup_name: Pure Sweet Sugars
Enter value for email: sales@puresweet.com
Enter value for phone: 9654321098
Enter value for address: 88 Sugar Mill Road, Bengaluru
old 2: VALUES ('&sup_id', '&sup_name', '&email', &phone, '&address')
new 2: VALUES ('SUP-003', 'Pure Sweet Sugars', 'sales@puresweet.com', 9654321098, '88 Sugar Mill Road, Bengaluru')

1 row created.
```

Interactive SQL Insertion Template for Inventory Table

```
SQL> INSERT INTO Inventory (Inventory_ID, Product_ID, Supplier_ID, Stock_Quantity, Stock_Status, Last_Updated)
  2 VALUES ('&inv_id', '&prod_id', '&sup_id', &qty, '&status', TO_DATE('&updated_date', 'YYYY-MM-DD'));
Enter value for inv_id: INV-001
Enter value for prod_id: BAK-101
Enter value for sup_id: SUP-001
Enter value for qty: 50
Enter value for status: Available
Enter value for updated_date: 2026-03-15
old 2: VALUES ('&inv_id', '&prod_id', '&sup_id', &qty, '&status', TO_DATE('&updated_date', 'YYYY-MM-DD'))
new 2: VALUES ('INV-001', 'BAK-101', 'SUP-001', 50, 'Available', TO_DATE('2026-03-15', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inv_id: INV-002
Enter value for prod_id: BAK-102
Enter value for sup_id: SUP-002
Enter value for qty: 15
Enter value for status: Available
Enter value for updated_date: 2026-03-16
old 2: VALUES ('&inv_id', '&prod_id', '&sup_id', &qty, '&status', TO_DATE('&updated_date', 'YYYY-MM-DD'))
new 2: VALUES ('INV-002', 'BAK-102', 'SUP-002', 15, 'Available', TO_DATE('2026-03-16', 'YYYY-MM-DD'))

1 row created.
```

```

SQL> /
Enter value for inv_id: INV-003
Enter value for prod_id: BAK-103
Enter value for sup_id: SUP-003
Enter value for qty: 0
Enter value for status: Out of Stock
Enter value for updated_date: 2026-03-10
old 2: VALUES ('&inv_id', '&prod_id', '&sup_id', &qty, '&status', TO_DATE('&updated_date', 'YYYY-MM-DD'))
new 2: VALUES ('INV-003', 'BAK-103', 'SUP-003', 0, 'Out of Stock', TO_DATE('2026-03-10', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for inv_id: INV-004
Enter value for prod_id: BAK-104
Enter value for sup_id: SUP-002
Enter value for qty: 30
Enter value for status: Available
Enter value for updated_date: 2026-03-17
old 2: VALUES ('&inv_id', '&prod_id', '&sup_id', &qty, '&status', TO_DATE('&updated_date', 'YYYY-MM-DD'))
new 2: VALUES ('INV-004', 'BAK-104', 'SUP-002', 30, 'Available', TO_DATE('2026-03-17', 'YYYY-MM-DD'))

1 row created.

```

Verification Display Queries

Query all records from Supplier table

```

SQL> SELECT * FROM Supplier;

```

SUPPLIE	SUPPLIER_NAME	EMAIL	PHONE

ADDRESS			

SUP-001	Golden Wheat Mills 12 Industrial Estate, Chennai	info@goldenwheat.com	9876543210
SUP-002	Dairy Fresh Farms 45 Green Valley, Coimbatore	contact@dairyfresh.com	9765432109
SUP-003	Pure Sweet Sugars 88 Sugar Mill Road, Bengaluru	sales@puresweet.com	9654321098

Query all records from Inventory table

```

SQL> SELECT * FROM Inventory;

```

INVENTO	PRODUCT	SUPPLIE	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
-----	-----	-----	-----	-----	-----
INV-001	BAK-101	SUP-001	50	Available	15-MAR-26
INV-002	BAK-102	SUP-002	15	Available	16-MAR-26
INV-003	BAK-103	SUP-003	0	Out of Stock	10-MAR-26
INV-004	BAK-104	SUP-002	30	Available	17-MAR-26

5. Supplier and Inventory Analytical Reports

Report 1: Supplier-Wise Available Stock Summary

```
SQL> SELECT s.Supplier_ID,
2         s.Supplier_Name,
3         s.Phone,
4         SUM(i.Stock_Quantity) AS Total_Stock_Units
5 FROM Supplier s
6 JOIN Inventory i ON s.Supplier_ID = i.Supplier_ID
7 GROUP BY s.Supplier_ID, s.Supplier_Name, s.Phone;
```

SUPPLIE	SUPPLIER_NAME	PHONE	TOTAL_STOCK_UNITS
SUP-002	Dairy Fresh Farms	9765432109	45
SUP-003	Pure Sweet Sugars	9654321098	0
SUP-001	Golden Wheat Mills	9876543210	50

Report 2: Stock Availability & Item Alert Report

```
SQL> SELECT i.Inventory_ID,
2         p.Title AS Product_Name,
3         s.Supplier_Name,
4         i.Stock_Quantity,
5         i.Stock_Status,
6         CASE
7             WHEN i.Stock_Quantity = 0 THEN 'CRITICAL: Reorder Required'
8             WHEN i.Stock_Quantity < 20 THEN 'WARNING: Low Stock'
9             ELSE 'OK: Sufficient Stock'
10        END AS Operational_Alert
11 FROM Inventory i
12 JOIN Product p ON i.Product_ID = p.Product_ID
13 JOIN Supplier s ON i.Supplier_ID = s.Supplier_ID;
```

INVENTO

PRODUCT_NAME

SUPPLIER_NAME STOCK_QUANTITY STOCK_STATUS

OPERATIONAL_ALERT

INV-001

Sourdough Bread

Golden Wheat Mills

50 Available

OK: Sufficient Stock

INVENTO

PRODUCT_NAME

SUPPLIER_NAME STOCK_QUANTITY STOCK_STATUS

OPERATIONAL_ALERT

INV-002

Chocolate Fudge Cake

Dairy Fresh Farms

15 Available

WARNING: Low Stock

```

INVENTO
-----
PRODUCT_NAME
-----
SUPPLIER_NAME          STOCK_QUANTITY STOCK_STATUS
-----
OPERATIONAL_ALERT
-----
INV-003
Almond Croissant
Pure Sweet Sugars          0 Out of Stock
CRITICAL: Reorder Required

INVENTO
-----
PRODUCT_NAME
-----
SUPPLIER_NAME          STOCK_QUANTITY STOCK_STATUS
-----
OPERATIONAL_ALERT
-----
INV-004
Red Velvet Cupcake
Dairy Fresh Farms          30 Available
OK: Sufficient Stock

```

Report 3: Inventory Metrics Analysis

3a. Total available stock items across inventory

```

SQL> SELECT SUM(Stock_Quantity) AS Aggregate_Available_Stock
2   FROM Inventory
3  WHERE Stock_Status = 'Available';

AGGREGATE_AVAILABLE_STOCK
-----
                        95

```

3b. Count of out-of-stock items

```

SQL> SELECT COUNT(*) AS Out_Of_Stock_Count
2   FROM Inventory
3  WHERE Stock_Status = 'Out of Stock';

OUT_OF_STOCK_COUNT
-----
                  1

```

3c. Highest stocked product details (using Subquery)

```
SQL> SELECT i.Inventory_ID, p.Title, i.Stock_Quantity
2  FROM Inventory i
3  JOIN Product p ON i.Product_ID = p.Product_ID
4  WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

INVENTO

TITLE

STOCK_QUANTITY

INV-001

Sourdough Bread

50

3d. Overall average stock quantity per batch

```
SQL> SELECT ROUND(AVG(Stock_Quantity), 1) AS Average_Stock_Per_Batch
2  FROM Inventory;
```

AVERAGE_STOCK_PER_BATCH

23.8